
Modern Signal Analysis and Data Processing

Name: Haozheng Ji
Student Number: 12311405

Email: jihz2023@mail.sustech.edu.cn
Assignment: Number 5

Problem 1 Show that the convolution of two continuous functions $f(t)$ and $g(t)$ is equivalent to the multiplication of their Fourier transforms, i.e.,

$$\mathcal{F}\{f * g\} = \mathcal{F}\{f\} \cdot \mathcal{F}\{g\}.$$

The Fourier transform converts a function from the time domain into the frequency domain. One of its most important properties is that convolution in the time domain becomes multiplication in the frequency domain.

The convolution of $f(t)$ and $g(t)$ is defined as

$$c(t) = (f * g)(t) = \int_{-\infty}^{\infty} f(t - \tau)g(\tau) d\tau.$$

Taking the Fourier transform of $c(t)$,

$$\mathcal{F}_c(\omega) = \int_{-\infty}^{\infty} c(t)e^{-j\omega t} dt.$$

Substituting the definition of convolution,

$$\mathcal{F}_c(\omega) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t - \tau)g(\tau)e^{-j\omega t} d\tau dt.$$

Let

$$\psi = t - \tau, \quad t = \psi + \tau,$$

then

$$\begin{aligned} \mathcal{F}_c(\omega) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(\psi)g(\tau)e^{-j\omega(\psi+\tau)} d\tau d\psi. \\ &= \left(\int_{-\infty}^{\infty} f(\psi)e^{-j\omega\psi} d\psi \right) \left(\int_{-\infty}^{\infty} g(\tau)e^{-j\omega\tau} d\tau \right). \end{aligned}$$

Recognizing these integrals as the Fourier transforms of f and g ,

$$\mathcal{F}_c(\omega) = \mathcal{F}_f(\omega)\mathcal{F}_g(\omega).$$

Therefore,

$$\boxed{\mathcal{F}\{f * g\} = \mathcal{F}\{f\} \cdot \mathcal{F}\{g\}}.$$

Problem 2 Show that the discrete convolution of two sequences $f[n]$ and $g[n]$ is equivalent to the multiplication of their discrete Fourier transforms (DFT), i.e.,

$$\mathcal{F}\{f * g\} = \mathcal{F}\{f\} \cdot \mathcal{F}\{g\}.$$

The discrete Fourier transform (DFT) converts a finite discrete sequence into its frequency-domain representation.

The DFTs of $f[n]$ and $g[n]$ are defined as

$$\mathcal{F}_f(n) = \sum_{k=0}^{N-1} f[k] e^{-j \frac{2\pi}{N} kn},$$

$$\mathcal{F}_g(n) = \sum_{k=0}^{N-1} g[k] e^{-j \frac{2\pi}{N} kn}.$$

The discrete convolution is

$$(f * g)[k] = \sum_{m=0}^{N-1} f[m] g[k - m].$$

Taking the DFT of the convolution,

$$\mathcal{F}_{f*g}(n) = \sum_{k=0}^{N-1} (f * g)[k] e^{-j \frac{2\pi}{N} kn} = \sum_{k=0}^{N-1} \sum_{m=0}^{N-1} f[m] g[k - m] e^{-j \frac{2\pi}{N} kn}.$$

Rearranging the summations,

$$\mathcal{F}_{f*g}(n) = \sum_{m=0}^{N-1} f[m] \sum_{k=0}^{N-1} g[k - m] e^{-j \frac{2\pi}{N} kn}.$$

Let

$$\psi = k - m, \quad k = \psi + m.$$

Then,

$$\mathcal{F}_{f*g}(n) = \sum_{m=0}^{N-1} f[m] \sum_{\psi=-m}^{N-1-m} g[\psi] e^{-j \frac{2\pi}{N} (\psi+m)n} = \sum_{m=0}^{N-1} f[m] e^{-j \frac{2\pi}{N} mn} \sum_{\psi=-m}^{N-1-m} g[\psi] e^{-j \frac{2\pi}{N} \psi n}.$$

Because the sequence is periodic with period N ,

$$g[\psi + N] e^{-j \frac{2\pi}{N} (\psi+N)n} = g[\psi] e^{-j \frac{2\pi}{N} \psi n}.$$

Therefore,

$$\begin{aligned} & \sum_{\psi=-m}^{N-1-m} g[\psi] e^{-j \frac{2\pi}{N} \psi n} \\ &= \sum_{\psi=-m}^{-1} g[\psi] e^{-j \frac{2\pi}{N} \psi n} + \sum_{\psi=0}^{N-1-m} g[\psi] e^{-j \frac{2\pi}{N} \psi n} = \sum_{\psi=-m+N}^{N-1} g[\psi] e^{-j \frac{2\pi}{N} \psi n} + \sum_{\psi=0}^{N-1-m} g[\psi] e^{-j \frac{2\pi}{N} \psi n} \\ &= \sum_{\psi=0}^{N-1-m} g[\psi] e^{-j \frac{2\pi}{N} \psi n} + \sum_{\psi=-m+N}^{N-1} g[\psi] e^{-j \frac{2\pi}{N} \psi n} = \sum_{\psi=0}^{N-1} g[\psi] e^{-j \frac{2\pi}{N} \psi n} = \mathcal{F}_g(n). \end{aligned}$$

Substituting this result,

$$\mathcal{F}_{f*g}(n) = \sum_{m=0}^{N-1} f[m] \mathcal{F}_g(n) e^{-j \frac{2\pi}{N} mn} = \mathcal{F}_g(n) \sum_{m=0}^{N-1} f[m] e^{-j \frac{2\pi}{N} mn}.$$

Recognizing the remaining summation as $\mathcal{F}_f(n)$,

$$\mathcal{F}_{f*g}(n) = \mathcal{F}_f(n) \mathcal{F}_g(n).$$

Hence,

$$\boxed{\mathcal{F}\{f * g\} = \mathcal{F}\{f\} \cdot \mathcal{F}\{g\}}.$$

Problem 3 Use the forward and inverse fourier transform to calculate the convolution of discrete sequences, and use directly calculating the convolution to compare the time cost of the two methods.

As we can see, as the length of the sequence increases, the time cost of two methods both increase, but the time cost of direct convolution increases much faster than the method using Fourier transform. This is because the direct convolution has a time complexity of $O(N^2)$, while the method using Fourier transform has a time complexity of $O(N \log N)$ due to the use of Fast Fourier Transform (FFT) algorithms.

